

R1

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2

R1(config)#ip route 15.0.0.0 255.0.0.0 20.0.0.2

R1(config)#ip route 21.0.0.0 255.0.0.0 20.0.0.2

R1(config)#ip route 172.19.0.0 255.255.0.0 20.0.0.2

% Invalid input detected at '^' marker.

R1(config)#no ip route 17.19.0.0 255.255.0.0 20.0.0.2

%No matching route to delete

R1(config)#ip route 172.19.0.0 255.255.0.0 20.0.0.2

R1(config)#ip route 192.168.168.0 255.255.255.0 20.0.0.2

R1(config)#ip route 162.14.0.0 255.255.0.0 20.0.0.2

R1(config)#end

R1#

*Jul 21 00:52:53.825: %SYS-5-CONFIG_I: Configured from console by console

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.0.0.0/8 is directly connected, Ethernet0/0

L 10.0.0.1/32 is directly connected, Ethernet0/0

S 15.0.0.0/8 [1/0] via 20.0.0.2

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 20.0.0.0/8 is directly connected, Ethernet0/1

L 20.0.0.1/32 is directly connected, Ethernet0/1

S 21.0.0.0/8 [1/0] via 20.0.0.2

S 30.0.0.0/8 [1/0] via 20.0.0.2

S 162.14.0.0/16 [1/0] via 20.0.0.2

S 172.19.0.0/16 [1/0] via 20.0.0.2

S 192.168.168.0/24 [1/0] via 20.0.0.2

R1#

R1#

 R2

```
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 21.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
      ^
% Invalid input detected at '^' marker.

R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
*Jul 21 00:58:41.645: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#end
R2#s
*Jul 21 01:00:13.477: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S      10.0.0.0/8 [1/0] via 20.0.0.1
S      15.0.0.0/8 [1/0] via 30.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      20.0.0.0/8 is directly connected, Ethernet0/0
L      20.0.0.2/32 is directly connected, Ethernet0/0
S      21.0.0.0/8 [1/0] via 30.0.0.2
      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/1
L      30.0.0.1/32 is directly connected, Ethernet0/1
S      162.14.0.0/16 [1/0] via 30.0.0.2
S      172.19.0.0/16 [1/0] via 30.0.0.2
S      192.168.168.0/24 [1/0] via 30.0.0.2
R2#
R2#
```

```

C      172.19.0.0/16 is directly connected, Ethernet0/1
--More--
*Jul 21 01:05:33.911: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
L      172.19.0.1/32 is directly connected, Ethernet0/1
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
%Invalid next hop address (it's this router)
R3(config)#ip route 15.0.0.0 255.0.0.0 172.19.0.2
R3(config)#ip route 162.14.0.0 255.255.0.0 21.0.0.2
R3(config)#ip route 192.168.168.0 255.255.255.0 21.0.0.2
R3(config)#end
R3#sh ip r
*Jul 21 01:08:05.243: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S      10.0.0.0/8 [1/0] via 30.0.0.1
S      15.0.0.0/8 [1/0] via 172.19.0.2
S      20.0.0.0/8 [1/0] via 30.0.0.1
S      21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
S      30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
S      162.14.0.0/16 [1/0] via 21.0.0.2
S      172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/1
L      172.19.0.1/32 is directly connected, Ethernet0/1
S      192.168.168.0/24 [1/0] via 21.0.0.2

```


R4

Ethernet0/0	15.0.0.1	YES	manual	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R4#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1

R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1

R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1

R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1

R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1

R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1

R4(config)#end

R4#sh ip r

*Jul 21 01:12:52.540: %SYS-5-CONFIG_I: Configured from console by console

R4#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 172.19.0.1

15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 15.0.0.0/8 is directly connected, Ethernet0/0

L 15.0.0.1/32 is directly connected, Ethernet0/0

S 20.0.0.0/8 [1/0] via 172.19.0.1

S 21.0.0.0/8 [1/0] via 172.19.0.1

S 30.0.0.0/8 [1/0] via 172.19.0.1

S 162.14.0.0/16 [1/0] via 172.19.0.1

172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks

C 172.19.0.0/16 is directly connected, Ethernet0/1

--More--

*Jul 21 01:12:58.498: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

L 172.19.0.2/32 is directly connected, Ethernet0/1

S 192.168.168.0/24 [1/0] via 172.19.0.1

R4#

R5

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*Jul 21 01:16:04.431: %SYS-5-CONFIG_I: Configured from console by console
R5# sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              21.0.0.2        YES manual up          up
Ethernet0/1              192.168.168.1   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#ip route 10.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 20.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 30.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 15.0.0.0 255.0.0.0 21.0.0.1
R5(config)#ip route 172.19.0.0 255.255.0.0 21.0.0.1
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
R5(config)#end
R5#sh ip r
*Jul 21 01:19:43.056: %SYS-5-CONFIG_I: Configured from console by console
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     21.0.0.0/8 is directly connected, Ethernet0/0
L     21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
```

R6

```
L      162.14.0.1/32 is directly connected, Ethernet0/0
S      172.19.0.0/16 [1/0] via 192.168.168.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.168.0/24 is directly connected, Ethernet0/1
L      192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
R6#
R6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#ip route 21.0.0.0 255.0.0.0 192.168.168.1
R6(config)#end
R6#s
*Jul 21 01:26:23.659: %SYS-5-CONFIG_I: Configured from console by console
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S      10.0.0.0/8 [1/0] via 192.168.168.1
S      15.0.0.0/8 [1/0] via 192.168.168.1
S      20.0.0.0/8 [1/0] via 192.168.168.1
S      21.0.0.0/8 [1/0] via 192.168.168.1
S      30.0.0.0/8 [1/0] via 192.168.168.1
      162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      162.14.0.0/16 is directly connected, Ethernet0/0
L      162.14.0.1/32 is directly connected, Ethernet0/0
S      172.19.0.0/16 [1/0] via 192.168.168.1
      192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.168.0/24 is directly connected, Ethernet0/1
L      192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
R6#
```



```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 162.14.0.1
Pinging 162.14.0.1 with 32 bytes of data:
Reply from 162.14.0.1: bytes=32 time=3ms TTL=251
Reply from 162.14.0.1: bytes=32 time=2ms TTL=251
Reply from 162.14.0.1: bytes=32 time=2ms TTL=251
Reply from 162.14.0.1: bytes=32 time=2ms TTL=251
Ping statistics for 162.14.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms
C:\Documents and Settings\user>ping 162.14.0.10
Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\Documents and Settings\user>
C:\Documents and Settings\user>
```

```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe

Pinging 192.168.168.2 with 32 bytes of data:
Reply from 192.168.168.2: bytes=32 time=2ms TTL=252
Reply from 192.168.168.2: bytes=32 time=2ms TTL=252
Reply from 192.168.168.2: bytes=32 time=2ms TTL=252
Reply from 192.168.168.2: bytes=32 time=2ms TTL=252

Ping statistics for 192.168.168.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_
```



```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\Documents and Settings\user>ping 15.0.0.10
Pinging 15.0.0.10 with 32 bytes of data:
Reply from 15.0.0.10: bytes=32 time=6ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=4ms TTL=124
Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 3ms
C:\Documents and Settings\user>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=4ms TTL=123
Reply from 10.0.0.10: bytes=32 time=4ms TTL=123
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms
C:\Documents and Settings\user>
```